

Avantis - APIs

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Response

Field Reference

Top-level

portfolio[i]

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Examples

cURL

JavaScript (fetch)

Common Calculations (client-side examples)

Status Codes

1.) Socket API – Pairs & Groups (Realtime + Snapshot)

API that returns the **current params/state** of all **pairs** and **groups**, plus a Socket.IO feed for realtime updates.

HTTP Snapshot Endpoint

```
GET https://socket-api-pub.avantisfi.com/socket-api/v1/data
```

Socket.IO Realtime

```
https://socket-api.avantisfi.com/socket.io
```

Description

- Returns a consolidated snapshot of group and pair configuration/state as derived from contracts.
- **overrides** apply on top of contract state (e.g., temporarily change a param or delist a pair).

- Socket.IO pushes **incremental changes** as they happen.

Auth

- None required (public endpoints).
-

Response (Snapshot)

Shape

```
{ "data": { "dataVersion": 1.5, "groupInfo": { "0": { "groupMaxOI": 28271089.47, "groupOI": 14424987.62, "name": "CRYPTO01", "maxOpenInterestP": 28, "isSpreadDynamic": true }, "1": { "groupMaxOI": 13125862.97, "groupOI": 5361155.26, "name": "CRYPTO02", "maxOpenInterestP": 13, "isSpreadDynamic": true } }, "pairInfos": { "0": { "feed": { "maxOpenDeviationP": 1, "maxCloseDeviationP": 5, "feedId": "0x...", "attributes": { "symbol": "Crypto.ETH/USD", "asset_type": "Crypto", "is_open": true, "next_open": 0, "next_close": 0, "schedule": "America/New_York;0,0,0,0,0,0,0;" } }, "backupFeed": { "maxDeviationP": 1.5, "feedId": "0x..." }, "spreadP": 0.01, "pnlSpreadP": 0.01, "leverages": { "minLeverage": 1, "maxLeverage": 75, "pnlMinLeverage": 75, "pnlMaxLeverage": 500 }, "priceImpactMultiplier": 1.2, "skewImpactMultiplier": 0.25, "groupIndex": 0, "feeIndex": 0, "values": { "maxGainP": 2500, "maxSLP": 80, "maxLongOiP": 50, "maxShortOiP": 50, "groupOpenInterestPercentageP": 100, "maxWalletOI": 50, "isUSDCAligned": false }, "from": "ETH", "to": "USD", "timer": { "numTiers": 1, "positionSizeToThresholdTierMap": { "0": 0 }, "thresholdTierToTimerMap": { "0": 0 } }, "openInterest": { "long": 3479113.63, "short": 3222008.27 }, "marginFee": { "long": 0.0017123287671232876, "short": 0.0017123287671232876 }, "pairOI": 6701121.9, "pairMaxOI": 28271089.47, "maxWalletOI": 45435679.51, "pairParams": { "onePercentDepthAbove": 19719825.472, "onePercentDepthBelow": 27371509.912 }, "blockOILimit": 2500000, "storagePairParams": { "posSpreadCap": 2, "negSpreadCap": 25, "isPnlTypeAllowed": 1, "pnlPriceImpactMultiplier": 1.2, "pnlSkewImpactMultiplier": 0, "pnlPosSpreadCap": 2, "pnlNegSpreadCap": 5, "minBorrowFee": 15 }, "openFeeP": 0.045, "closeFeeP": 0.045, "limitOrderFeeP": 0.01, "minLevPosUSDC": 10, "pnlFees": { "numTiers": 10, "tierP": [1, 5, 25, 50, 100, 250, 500, 1500, 2500, 3000], "feesP": [80, 50, 45, 37.5, 27.5, 25, 25, 22.5, 15, 2.5] }, "lossProtectionMultiplier": { "0": 0, "1": 8, "2": 8, "3": 8 }, "skewEqParams": [ [0, 450], [0, 450], [0, 450], [0, 450], [0, 450], [0, 450], [0, 450], [0, 450], [0, 450], [0, 450] ], "longSkewConfig": { "0": 0, "1": 50, "2": 70, "3": 80 }, "shortSkewConfig": { "0": 0, "1": 50, "2": 70, "3": 80 }, "isPairListed": true, "index": 0, "pairSpreadP": 0.01, "pairLimitOrderFeeP": 0.01, "pairMinLevPosUSDC": 10, "pairMinLeverage": 1 } }, "pairCount": 90, "maxTradesPerPair": 40, "totalOI": 38934218.65, "maxOpenInterest": 90871359.02, "overrides": { "pairInfos": { "52": { "leverages": { "maxLeverage": 50 } }, "65": { "from": "WTI", "to": "USD" }, "78": { "isPairListed": false }, "79": { "isPairListed": false } } }, "success": true }
```

Field Reference

data

- `dataVersion` `number` – Version of the payload shape/content.
- `groupInfo` `object<groupIndex, Group>` – Map of **group index** → **group state**.
- `pairInfos` `object<pairIndex, Pair>` – Map of **pair index** → **pair state**.
- `pairCount` `number` – Total number of pairs in snapshot.
- `maxTradesPerPair` `number` – Max simultaneous trades per pair.
- `totalOi` `number` – Total open interest across all pairs.
- `maxOpenInterest` `number` – Global OI cap across all groups.
- `overrides` `object` – Values **overriding** contract state (apply after reading base values).

Group

- `groupMaxOI` `number` – Max OI cap for the group.
- `groupOI` `number` – Current group OI.
- `name` `string` – Group name (e.g., `CRYPT01`).
- `maxOpenInterestP` `number` – Group max OI percentage (policy metric).
- `isSpreadDynamic` `boolean` – Whether spread is dynamically adjusted for this group.

Pair

- `index` `number` – Pair index. **Use in contracts/APIs.**
- `groupIndex` `number` – Group association for the pair.
- `isPairListed` `boolean` – If `false` , **ignore** this pair.
- `from` / `to` `string` – Base/quote symbols.
- `spreadP` , `pairSpreadP` `number` – Spread percentage for market orders.
- `pnlSpreadP` `number` – Spread percentage for zero-fee perp (ZFP) path.

- `leverages` `object` –
 - `minLeverage` / `maxLeverage` – For **fixed-fee** order type.
 - `pnlMinLeverage` / `pnlMaxLeverage` – For **zero-fee perp (ZFP)** order type.
- `values` `object` – Key trading limits:
 - `maxGainP` – Max TP %.
 - `maxSLP` – Max SL %.
 - `maxLongOiP` , `maxShortOiP` – Side OI caps (% of pair or group policy).
 - `maxWalletOI` , `maxWalletOIP` – Wallet-level OI cap (absolute / percent).
 - `groupOpenInterestPercentageP` – Group OI % used for policy.
 - `isUSDCAligned` – Price/settlement alignment flag.
- `openInterest` `object` – `long` , `short` current OI.
- `pairOI` `number` – Total OI on the pair.
- `pairMaxOI` `number` – Max OI allowed on the pair.
- `maxWalletOI` `number` – Max OI allowed per wallet for this pair.
- `marginFee` `object` – Hourly margin fee % for `long` and `short` .
- `openFeeP` , `closeFeeP` , `limitOrderFeeP` `number` – Fee percentages.
- `minLevPosUSDC` / `pairMinLevPosUSDC` `number` – Minimum USDC for leveraged positions.
- `priceImpactMultiplier` , `skewImpactMultiplier` `number` – Price/skew impact controls.
- `pairParams.onePercentDepthAbove/Below` `number` – Liquidity depth for ±1%.
- `blockOILimit` `number` – OI cap for a single block.
- `storagePairParams` `object` – Additional pair policy (caps, pnl multipliers, borrow fee min, etc.).
- `pnlFees` `object` – ZFP performance fee schedule: `tierP[]` thresholds vs `feesP[]` .
- `lossProtectionMultiplier` `map<tier, percent>` – Tier → **loss protection** %.
- `longSkewConfig` / `shortSkewConfig` `map<tier, skew%>` – Tier trigger per skew.
- `skewEqParams` `array` – Skew equalization parameters.

- `timer object` – Tiered timer configuration for thresholds.
- `feed object` – Primary Pyth price feed config (see **Feeds** below).
- `backupFeed object` – Backup feed config.

Feeds

- `feed.feedId` – **Pyth price feed ID**. Use it to fetch latest or realtime prices (and price update data for contracts).
- `feed.attributes` – Metadata from Pyth (`symbol` , `asset_type` , market schedule fields).
 - `is_open boolean` – Current market-open flag.
 - `next_open / next_close unix` – Market schedule markers.
 - **Market-open logic:**
 - If `is_open == false` and `now < next_open ( if next_open > 0 )` → **market closed**.
 - If `is_open == true` OR (`now > next_open` AND `now < next_close`) → **market open**.

Overrides

- `overrides.pairInfos[<pairIndex>]` – Values that **override** the base pair state (e.g., `maxLeverage` , `from/to` , `isPairListed`).

Liquidity & OI Checks

- **Available pair liquidity (USDC):**
 - `pairAvail = pairMaxOI - pairOI`
- **Available group liquidity for the pair's group:**
 - `groupAvail = groupInfo[groupIndex].groupMaxOI - groupInfo[groupIndex].groupOI`
- **Effective available liquidity (what the UI should consider):**
 - `available = min(pairAvail, groupAvail)`

Skew tiers

- Long skew % = $\text{long} / (\text{long} + \text{short})$
- Short skew % = $\text{short} / (\text{long} + \text{short})$
- With configs like $\{0:0, 1:50, 2:70, 3:80\}$, hitting $\geq 80\%$ on one side $\rightarrow$ tier 3.

Examples

cURL (Snapshot)

```
curl -s "https://socket-api-pub.avantisfi.com/socket-api/v1/data"
```

JavaScript (fetch snapshot)

```
const res = await fetch("https://socket-api-pub.avantisfi.com/socket-api/v1/data"); const { data } = await res.json(); // Example: compute available liquidity for pair 0 const p = data.pairInfos[0]; const g = data.groupInfo[p.groupIndex]; const pairAvail = p.pairMaxOI - p.pairOI; const groupAvail = g.groupMaxOI - g.groupOI; const available = Math.min(pairAvail, groupAvail);
```

JavaScript (Socket.IO realtime)

```
import { io } from "socket.io-client"; // Socket.IO endpoint const socket = io("https://socket-api.avantisfi.com", { path: "/socket.io", transports: ["websocket"], }); socket.on("connect", () => { console.log("connected", socket.id); }); // Example: generic change stream (event name will match server-side emission) socket.on("RES:DATA", (payload) => { // payload may include changed pair/group fields and/or overrides console.log("update", payload); }); socket.on("disconnect", () => console.log("disconnected"));
```

Status Codes (Snapshot)

- `200` – Success with payload shown above.
 - `4xx/5xx` – Non-OK responses.
-

2.) User positions and open limit orders

API for fetching **open positions** and **limit orders** of a trader.

3) GET User Data

Endpoint

```
GET https://core.avantisfi.com/user-data?trader={trader}
```

Example

```
GET https://core.avantisfi.com/user-data?trader=0x8e1c4e0a7e85b2490f6d811824515d6fad3115a6
```

Description

Returns all **active positions** and **open limit orders** of a trader address.

Query Parameters

- `trader` `string` – EVM address of trader.

Auth

- None required (public endpoint).
-

Response

Shape

```
{ "positions": [ { "isOneCT": false, "liquidationPrice": "481278299403", "rolloverFee": "10908", "trader": "0x8E1c4e0a7e85b2490f6d811824515D6FAD3115A6", "pairIndex": 62, "index": 0, "buy": false, "collateral": "20000000000", "leverage": "100000000000", "openPrice": "443574692469", "sl": "479060667866", "tp": "222108938128", "lossProtection": "1", "openedAt": 1758710931, "isPnl": false} ], "limitOrders": [ { "isOneCT": false, "trader": "0x8E1c4e0a7e85b2490f6d811824515D6FAD3115A6", "pairIndex": 21, "index": 0, "collateral": "30000000", "positionSize": "3000000000", "buy": false, "price": "37600000000000", "leverage": "100000000000", "tp": "37528560000000", "sl": "37670000000000", "slippageP": "30000000000", "block": 35961058, "executionFee": "0", "liquidationPrice": "37919600000000" } ] }
```

Field Reference

Positions

Each entry represents an **active position**.

- `isOneCT` – ignore (internal field).
- `liquidationPrice` `string` – Liquidation price, **10 decimals precision**. Divide by 1e10.
- `rolloverFee` `string` – Current accrued margin fee, **6 decimals precision**.
- `trader` `string` – Trader address.
- `pairIndex` `number` – Pair index identifier.
- `index` `number` – Index of trade for this pair.
- `buy` `boolean` – Trade direction (`true` = long, `false` = short).
- `collateral` `string` – Collateral in trade, **6 decimals**.
- `leverage` `string` – Leverage, **10 decimals**.
- `openPrice` `string` – Open price, **10 decimals**.
- `sl` `string` – Stop-loss price, **10 decimals** (0 if not set).
- `tp` `string` – Take-profit price, **10 decimals** (0 if not set).
- `lossProtection` `string` – Loss protection tier (`0` = none).

- `openedAt` `number` – Unix timestamp when trade was opened.
 - `isPnl` `boolean` – `true` if zero-fee perp (ZFP), `false` if fixed-fee trade.
-

Limit Orders

Each entry represents an **open order to open a trade**.

- Fields are same as **Positions** except:
 - `slippageP` `string` – Slippage percentage (10 decimals).
 - `block` `number` – Block number when limit order was registered.
 - `executionFee` `string` – Execution fee (currently `0`).
 - `liquidationPrice` `string` – Liquidation price at execution (10 decimals).
-

Examples

cURL

```
curl -s "https://core.avantisfi.com/user-data?trader=0x8e1c4e0a7e85b2490f6d811824515d6fad3115a6"
```

JavaScript (fetch)

```
const base = "https://core.avantisfi.com/user-data"; const trader = "0x8e1c4e0a7e85b2490f6d811824515d6fad3115a6"; const res = await fetch(`${base}?trader=${trader}`); const data = await res.json(); // Example: parse positions for (const pos of data.positions) { console.log({ pairIndex: pos.pairIndex, direction: pos.buy ? "LONG" : "SHORT", collateral: Number(pos.collateral) / 1e6, leverage: Number(pos.leverage) / 1e10, openPrice: Number(pos.openPrice) / 1e10, liqPrice: Number(pos.liquidationPrice) / 1e10, sl: Number(pos.sl) / 1e10, tp: Number(pos.tp) / 1e10, rolloverFee: Number(pos.rolloverFee) / 1e6, lossProtectionTier: Number(pos.lossProtection), openedAt: new Date(pos.openedAt * 1000), isZeroFeePerp: pos.isPnl }); }
```

Status Codes

- `200` – Success with positions/limitOrders.
- `4xx/5xx` – Error responses.

3.) Portfolio/Trades History

API for fetching **closed trades** for a specific user's portfolio.

1) GET Portfolio History

Endpoint

```
GET https://api.avantisfi.com/v2/history/portfolio/history/{userAddresses}/{pageNumber}
```

Example

```
GET https://api.avantisfi.com/v2/history/portfolio/history/0x8E1c4e0a7e85b2490f6d811824515D6FAD3115A6/1
```

Description

Returns a paginated list of all **closed trades** for the given `userAddress`.

Path Parameters

- `userAddress` `string` – EVM address (checksum or lowercase) whose portfolio history will be fetched.
- `pageNumber` `integer` – 1-based page index.

Auth

- None required (public endpoint).

Pagination

- `count` `number` – total number of items across all pages.
 - `pageCount` `number` – total number of pages.
 - `portfolio.length` per page may vary.
-

Response

Shape

```
{ "portfolio": [ { "_id": "68d20becf00c296dc8522c8d", "event": { "args": { "t": { "index": 1, "initialPosToken": 0.98542, "leverage": 36, "openPrice": 113983.22122069, "pairIndex": 1, "positionSizeUSDC": 1757.600519, "sl": 0, "tp": 193090.9, "trader": "0xBc100e94cF6B0655e6eabb70F3b443b862D18397", "buy": true, "timestamp": 1757600519 }, "price": 112002.68, "positionSizeUSDC": 0.98542, "usdcSentToTrader": 0, "isPnl": false, "_feeInfo": { "closingFee": 0.147813, "r": 0.222313, "lossProtectionPSum": 0, "lossProtection": 0, "liquidationFee": 0.147813, "keeperFee": 0, "actualCloseFee": 0 } } }, "_grossPnl": -0.615294, "timeStamp": "2025-09-23T02:54:35.000Z" } ], "count": 70892, "pageCount": 7090, "success": true }
```

Field Reference

Top-level

- `portfolio` `array<object>` – Closed trade entries.
- `count` `number` – Total items across all pages.
- `pageCount` `number` – Total pages available.
- `success` `boolean` – Request status.

`portfolio[i]`

- `_id` `string` – Internal identifier of the history record.
- `event` `object` – On-chain/event payload.

- `_grossPnl` `number` – Gross PnL for this close in **USDC** units (negative if loss).
- `timeStamp` `string (ISO 8601)` – Close timestamp.

`portfolio[i].event.args (close context)`

- `price` `number` – Closing price used for the trade close.
- `positionSizeUSDC` `number` – **Collateral closed** in this event (not position size).
- `usdcSentToTrader` `number` – Amount of USDC returned to trader on close.
- `isPnl` `boolean` – Whether this was a **zero-fee** perp trade settlement pathway.
- `_feeInfo` `object` – Breakdown of closing/liq/keeper fees.

`portfolio[i].event.args (trade et open)`